

Álvaro López Álvarez

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LinkedIn, GitHub, Portfolio

Junior Software Engineer building Java backend fundamentals through a self-directed Spring Boot and Apache Kafka project (in progress), plus Kotlin experience from native Android development. Used to working independently and owning problems end to end, evidenced by an R&D role designing an autonomous agent that resolves data-access policy without supervision. Complements this with two full-stack roles at ITI (Go, React/TypeScript), Docker containerization, and CI/CD via GitHub Actions. A consistent pattern of learning new stacks independently and getting things done, from the Kafka-based ticketing platform to a from-scratch HTTP/1.1 server.

Core Competencies

- **Java & Kotlin Development**: Spring Boot, JPA/Hibernate, JUnit; Kotlin via native Android (MVVM) development
- **Event-Driven & Distributed Systems**: Apache Kafka, Docker, microservices, NATS JetStream
- **Backend Fundamentals**: REST API design, PostgreSQL, OAuth 2.0/JWT
- **Autonomous Delivery**: independent feature ownership, self-managed R&D projects, agentic/LLM tooling
- **Full-Stack Complement**: Golang, TypeScript/JavaScript (React)

Professional Experience

- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  R&D Developer — *Feb 2026–Jun 2026*
  - Implemented LLM-based improvements to a search engine
  - Built an autonomous multi-step agent converting natural-language questions into validated, read-only SQL queries against a PostgreSQL schema, including safety validation and error-driven retries, run independently with minimal day-to-day oversight
  - Managed scope and delivery priorities on a production-adjacent data-governance platform, prioritizing schema selection and safety validation first as the specification evolved
- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  Full-Stack Developer Intern — *Feb 2024–Jun 2024*
  - Developed and maintained a business-critical corporate personnel management application (Intranet)
  - Implemented new frontend features using React and TypeScript, improving internal usability
  - Collaborated with frontend and backend engineers on backend development, ensuring smooth integration between services via version-controlled Git workflows

Selected Projects

- **Kafka + Spring Boot Ticketing Platform** (in progress): Self-directed project building event-driven microservices (reservation, payment, notification, audit) with Spring Boot 3 and Apache Kafka to learn producer/consumer patterns and event-driven microservice design.
- **ValenMove**: Native Android app (Kotlin, MVVM) unifying Valencia's public transport into one interface; reverse-engineered public APIs for real-time data. Final Degree Project, grade 9/10.

- **English Learning API**: Spring Boot REST API with JPA entities, DTOs/mappers, Jakarta validation, centralized error handling, and randomized practice challenges with attempt tracking.
- **HTTP from TCP (Java)**: Java re-implementation of an HTTP/1.1 server directly on raw sockets, built to deepen Java fundamentals: I/O, concurrency via executors, JUnit testing.

Education

- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Master's Degree in Computer Engineering (MUIINF) *2024–2026*
  In progress (expected Sep 2026). Topics: distributed systems, high-performance computing, cybersecurity, embedded systems.
- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Grado (BSc) in Ingeniería Informática, Software Engineering track *2020–2024*
  GPA: 8.88/10. Thesis: "ValenMove - Unified Public Transport App" (grade 9/10).

Languages

- Spanish (native), Catalan (native), English (B2, confident conversational).

References

- Available upon request.